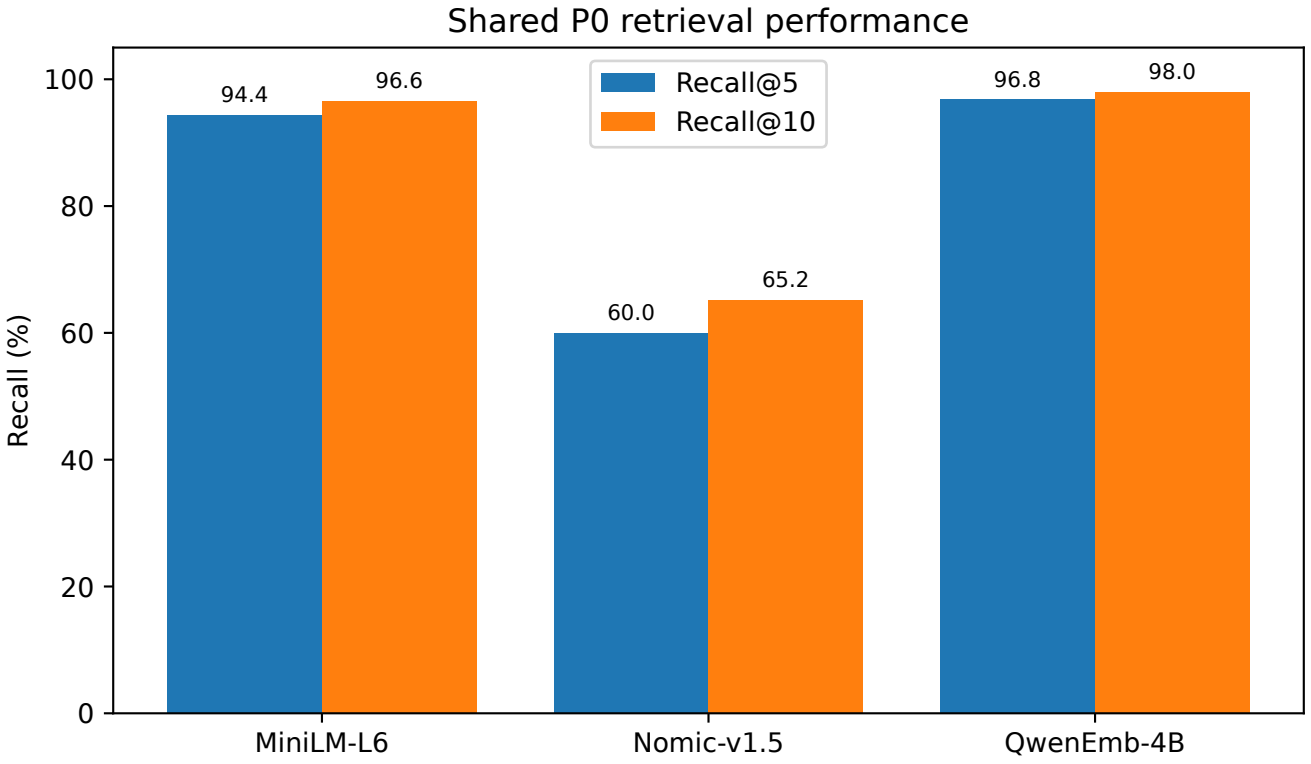
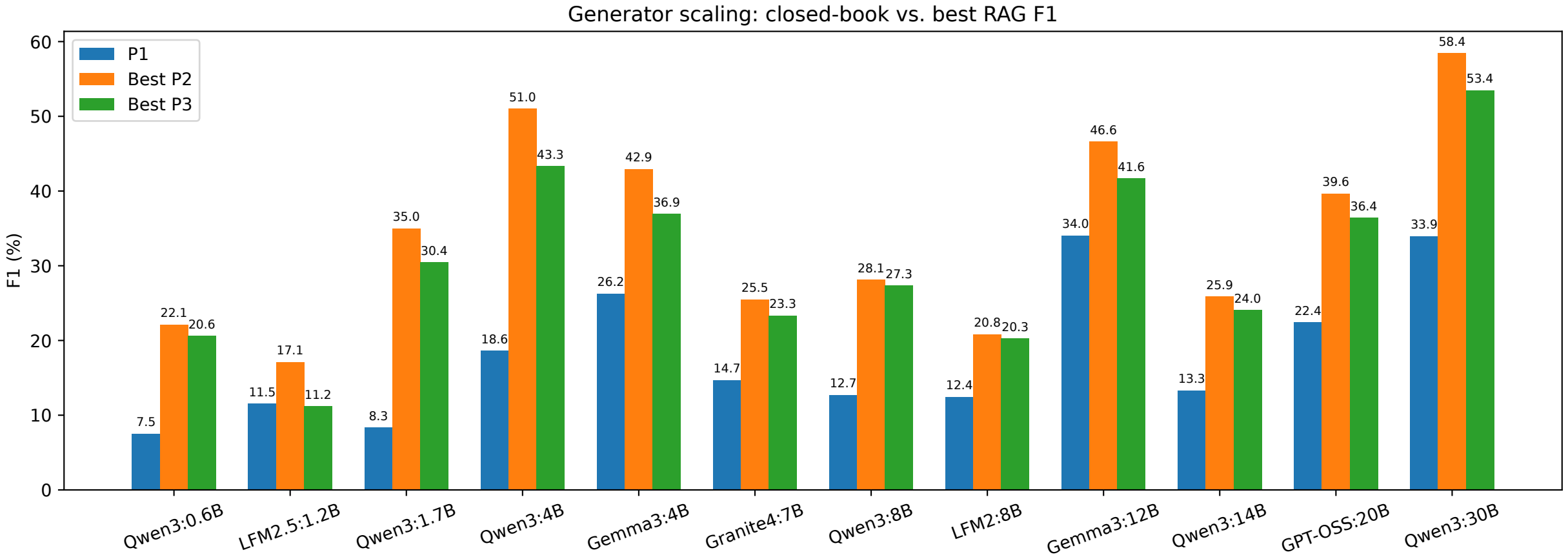
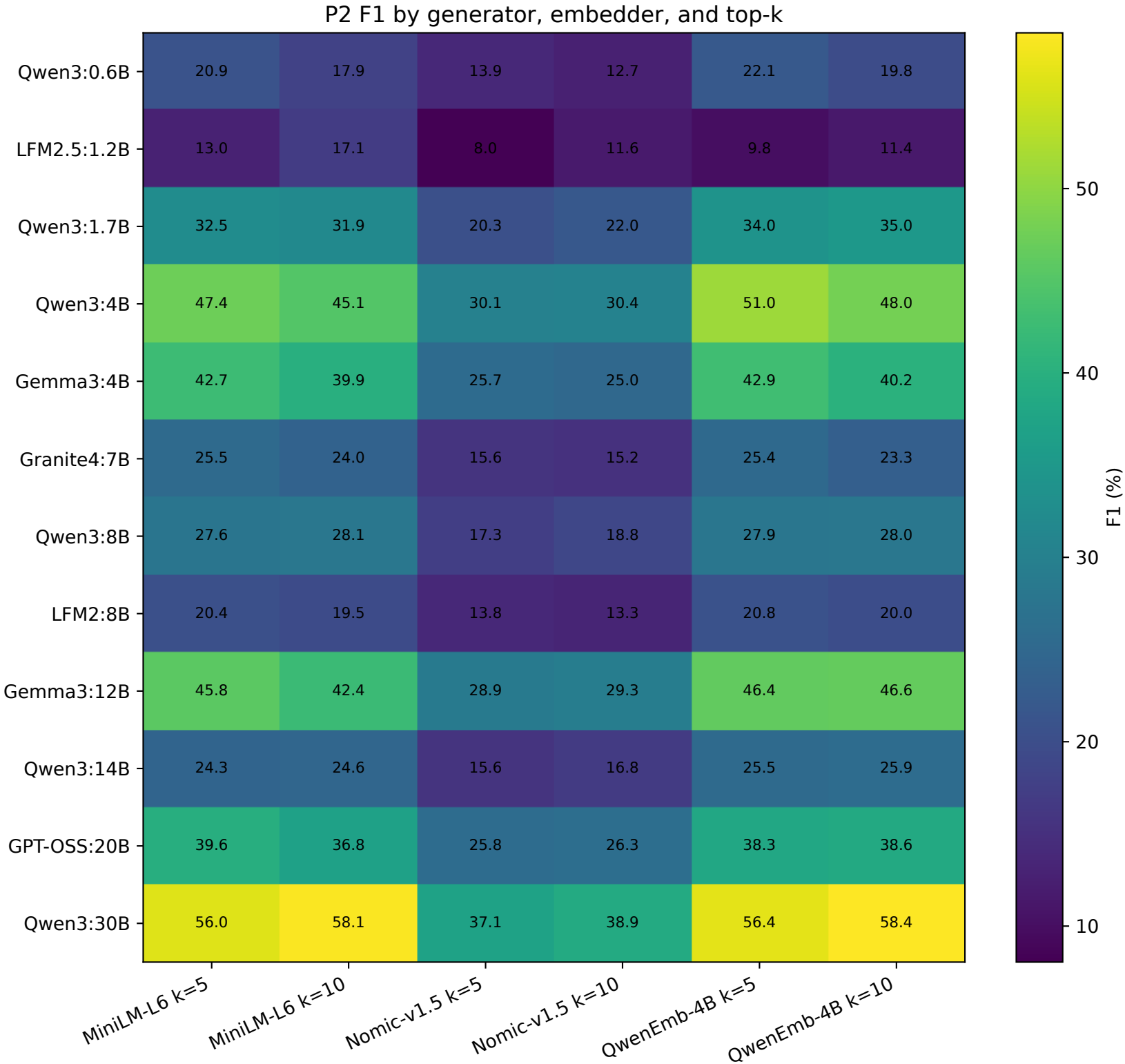
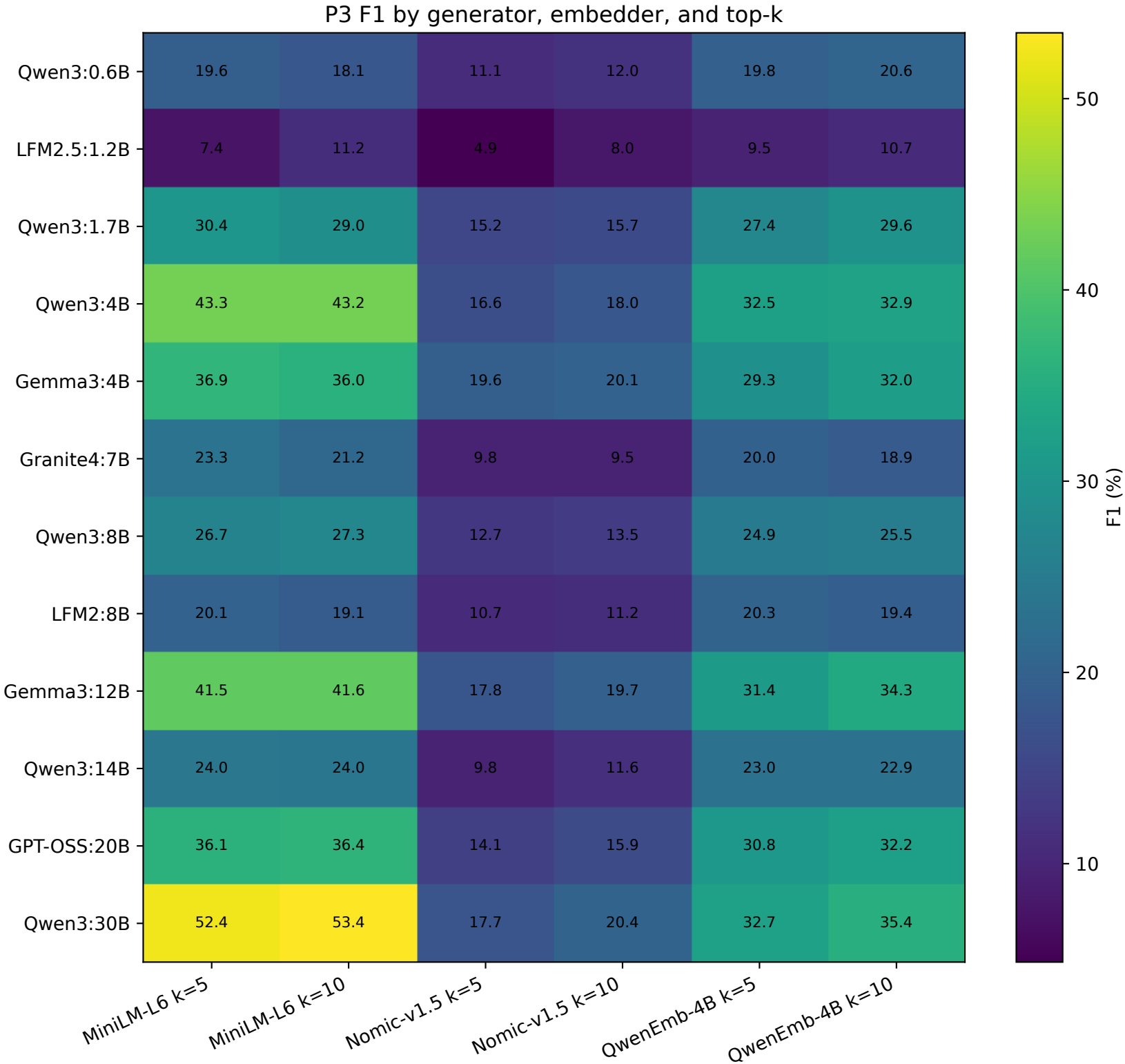
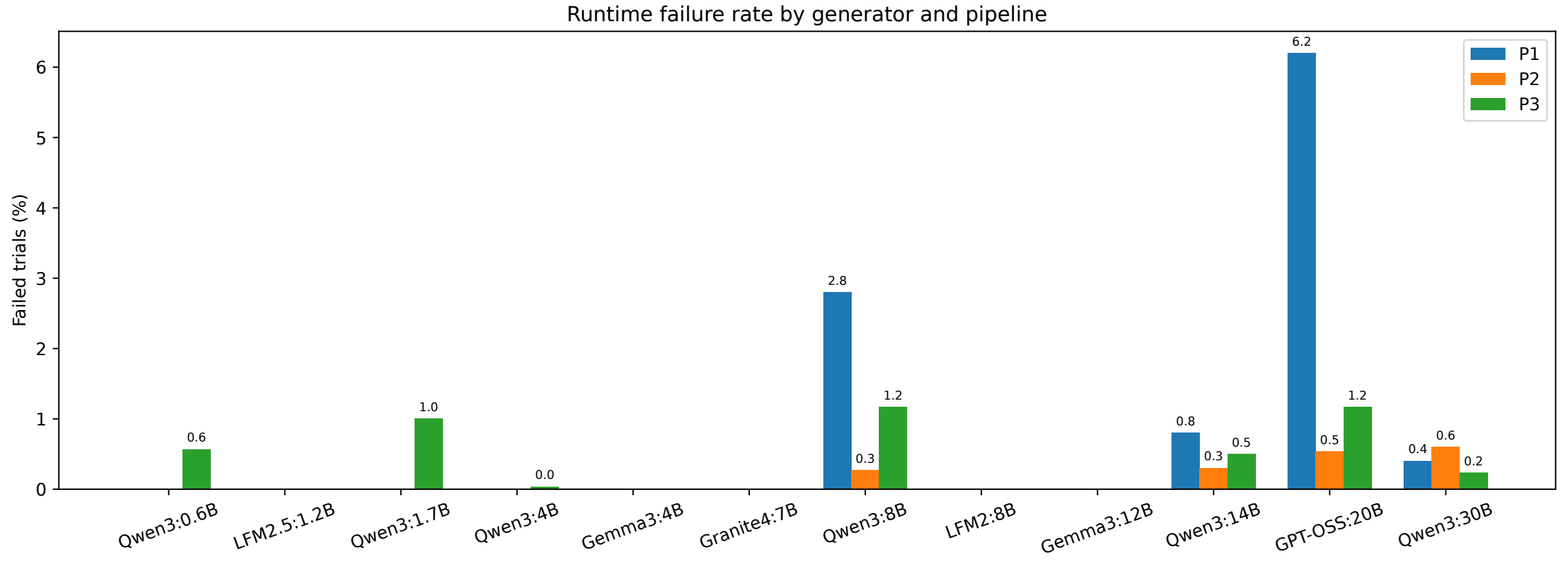
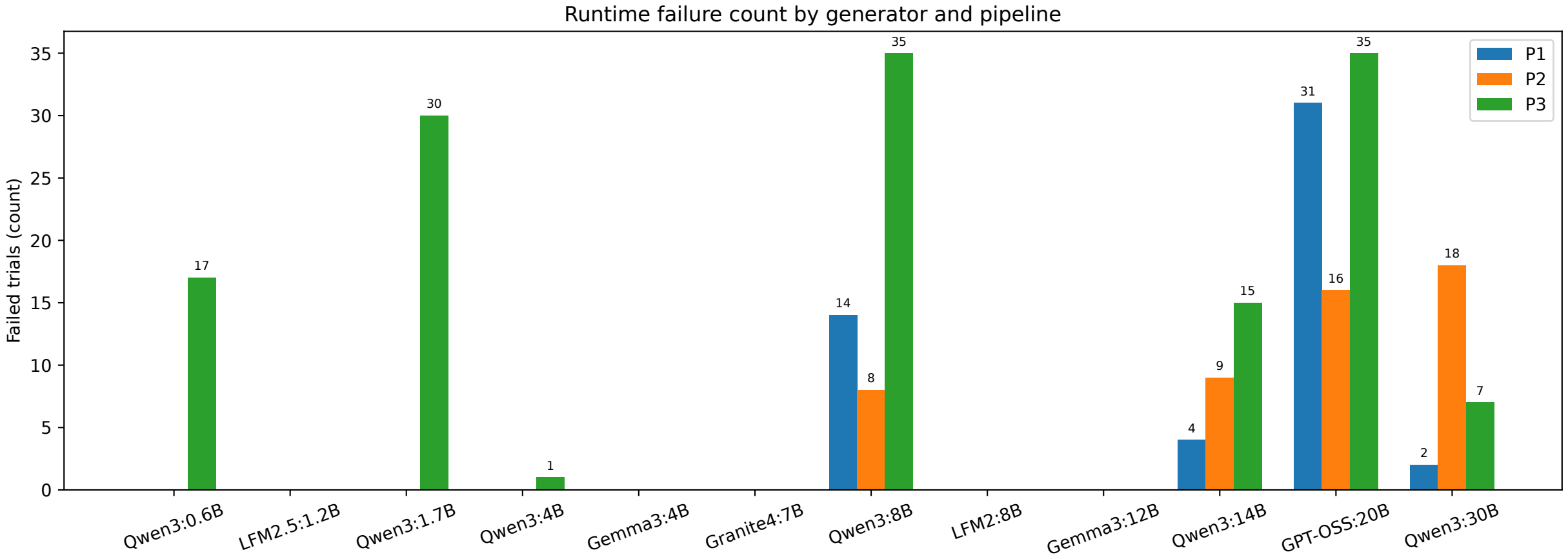

Shared P0 retrieval performance
Recall@5
Recall@10
Recall (%)
0
20
40
60
80
100
94.4
96.6
60.0
65.2
96.8
98.0
MiniLM-L6
Nomic-v1.5
QwenEmb-4B

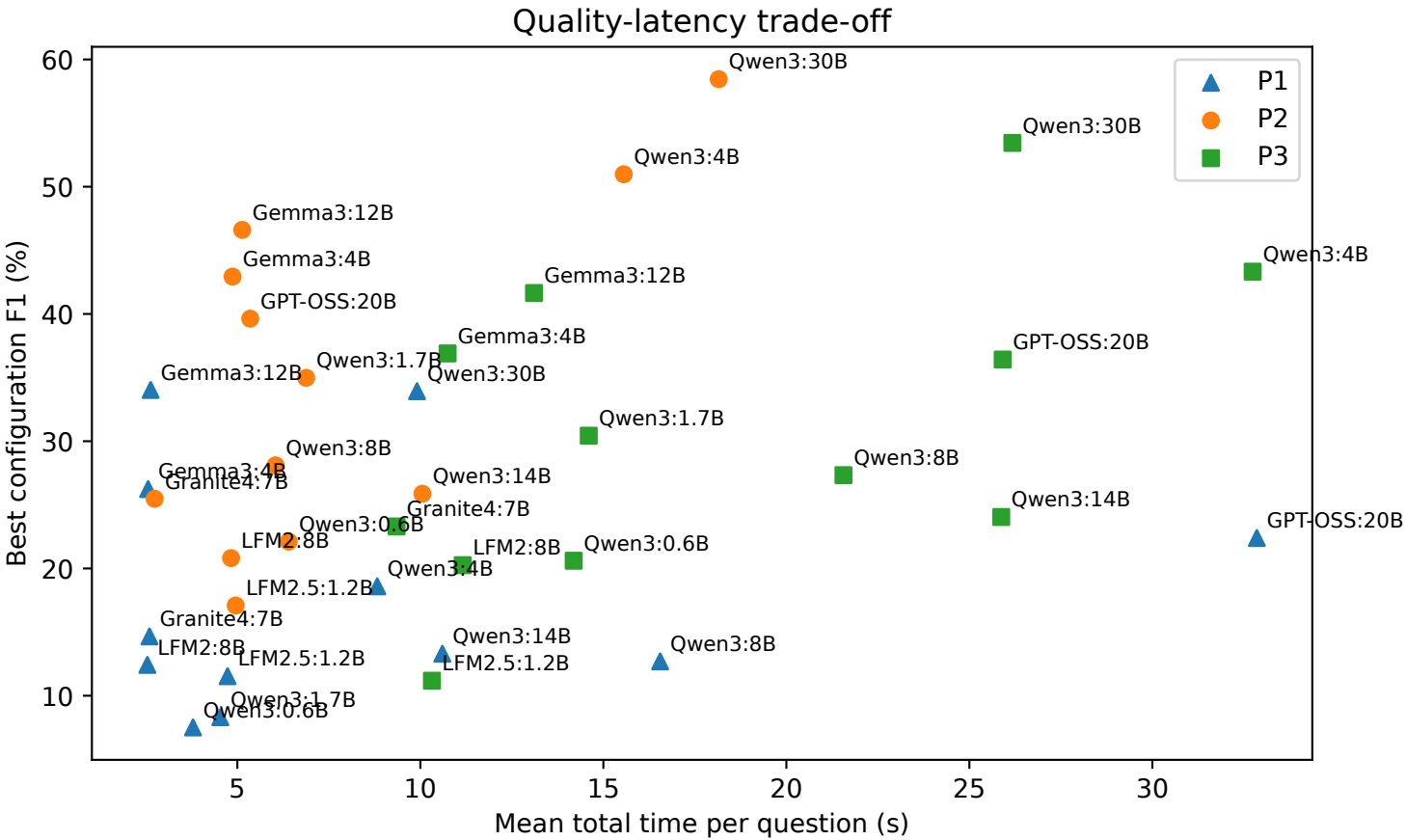

Generator scaling: closed-book vs. best RAG F1
P1
Best P2
Best P3
F1 (%)
0
10
20
30
40
50
60
7.5
22.1
20.6
11.5
17.1
11.2
8.3
35.0
30.4
18.6
51.0
43.3
26.2
42.9
36.9
14.7
25.5
23.3
12.7
28.1
27.3
12.4
20.8
20.3
34.0
46.6
41.6
13.3
25.9
24.0
22.4
39.6
36.4
33.9
58.4
53.4
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B

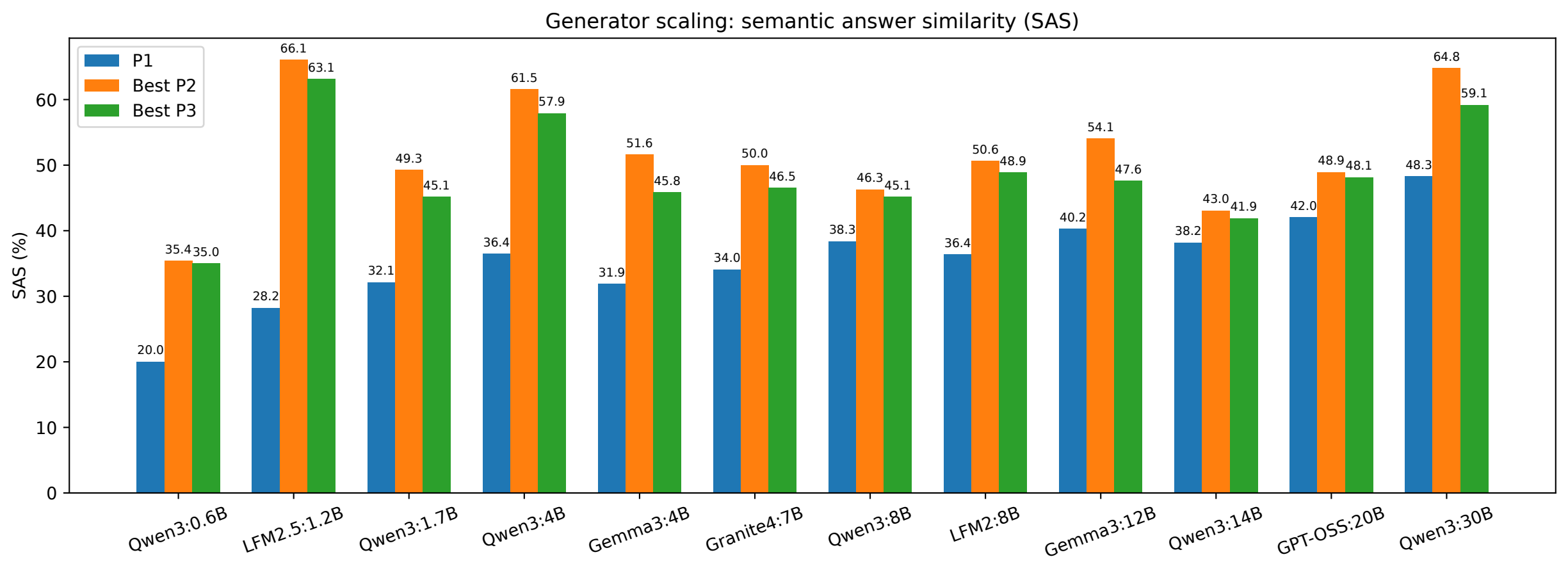

P2 F1 by generator, embedder, and top-k
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B
MiniLM-L6 k=5
MiniLM-L6 k=10
Nomic-v1.5 k=5
Nomic-v1.5 k=10
QwenEmb-4B k=5
QwenEmb-4B k=10
20.9 17.9 13.9 12.7 22.1 19.8
13.0 17.1 8.0 11.6 9.8 11.4
32.5 31.9 20.3 22.0 34.0 35.0
47.4 45.1 30.1 30.4 51.0 48.0
42.7 39.9 25.7 25.0 42.9 40.2
25.5 24.0 15.6 15.2 25.4 23.3
27.6 28.1 17.3 18.8 27.9 28.0
20.4 19.5 13.8 13.3 20.8 20.0
45.8 42.4 28.9 29.3 46.4 46.6
24.3 24.6 15.6 16.8 25.5 25.9
39.6 36.8 25.8 26.3 38.3 38.6
56.0 58.1 37.1 38.9 56.4 58.4
F1 (%)
10
20
30
40
50

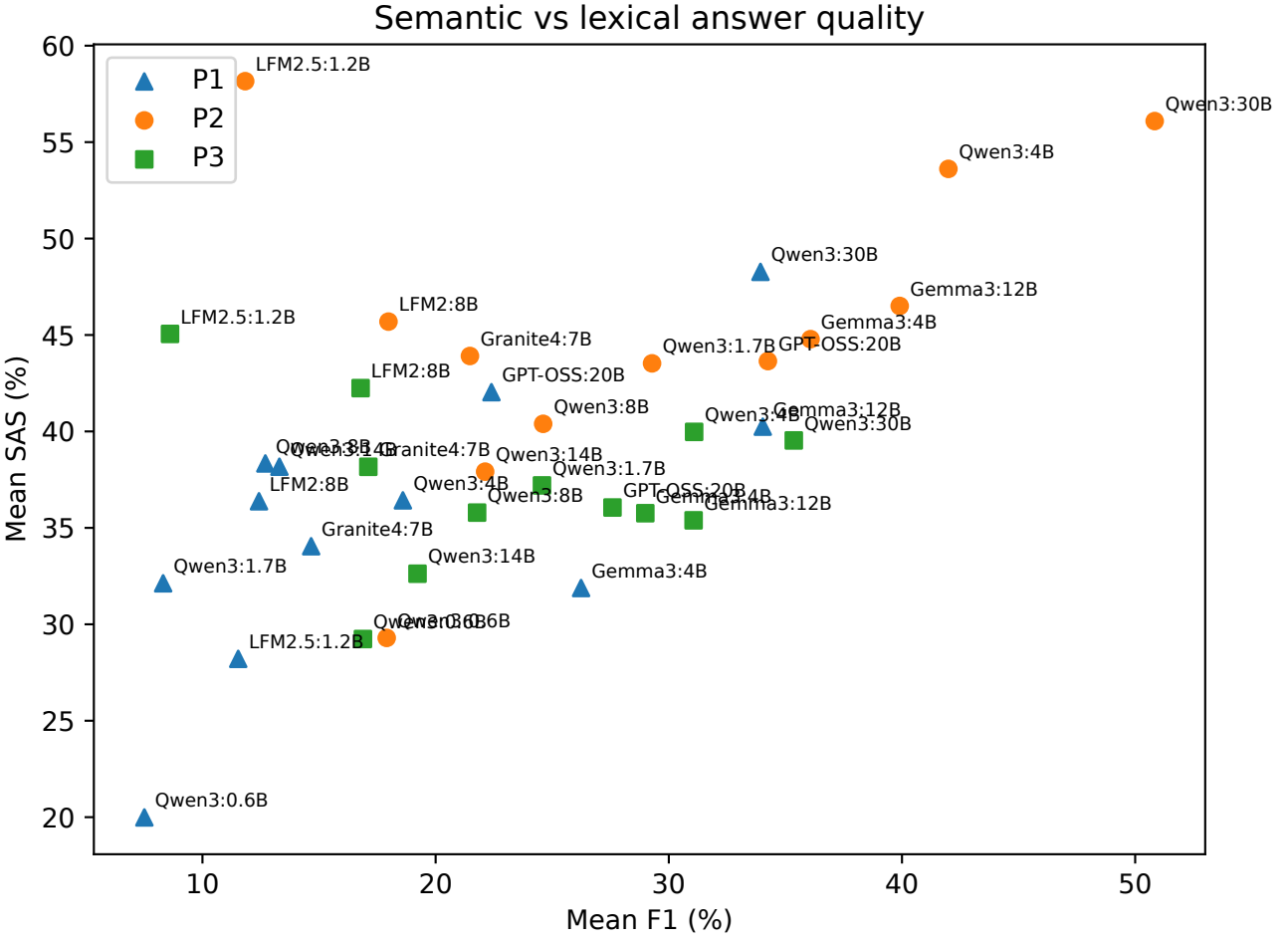

P3 F1 by generator, embedder, and top-k
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B
MiniLM-L6 k=5
MiniLM-L6 k=10
Nomic-v1.5 k=5
Nomic-v1.5 k=10
QwenEmb-4B k=5
QwenEmb-4B k=10
19.6 18.1 11.1 12.0 19.8 20.6
7.4 11.2 4.9 8.0 9.5 10.7
30.4 29.0 15.2 15.7 27.4 29.6
43.3 43.2 16.6 18.0 32.5 32.9
36.9 36.0 19.6 20.1 29.3 32.0
23.3 21.2 9.8 9.5 20.0 18.9
26.7 27.3 12.7 13.5 24.9 25.5
20.1 19.1 10.7 11.2 20.3 19.4
41.5 41.6 17.8 19.7 31.4 34.3
24.0 24.0 9.8 11.6 23.0 22.9
36.1 36.4 14.1 15.9 30.8 32.2
52.4 53.4 17.7 20.4 32.7 35.4
F1 (%)
50
40
30
20
10

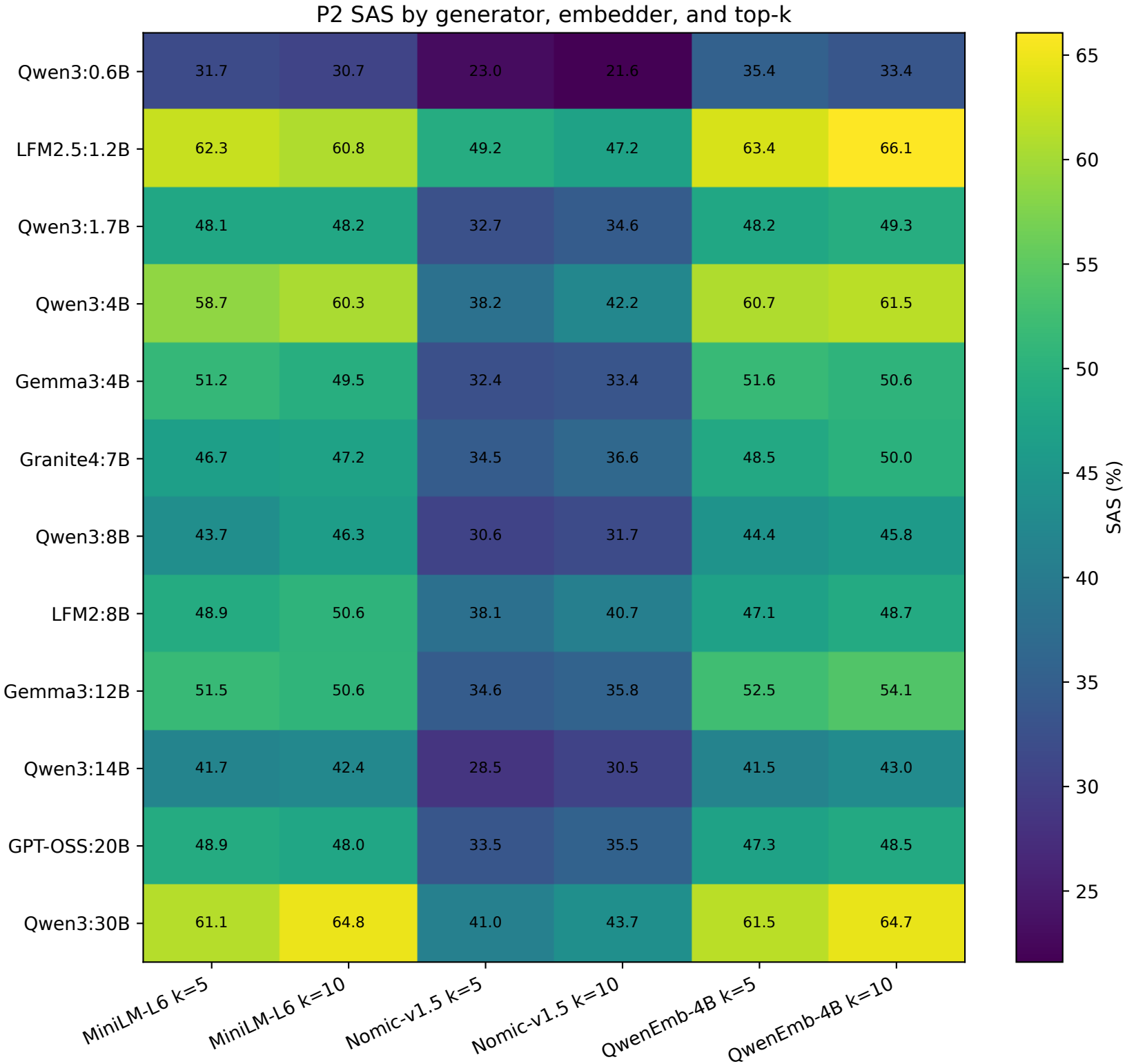

Runtime failure rate by generator and pipeline
Failed trials (%)
0
1
2
3
4
5
6
P1
P2
P3
0.6
1.0
0.0
2.8
0.3
1.2
0.8
0.3
0.5
6.2
0.5
1.2
0.4
0.6
0.2
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B

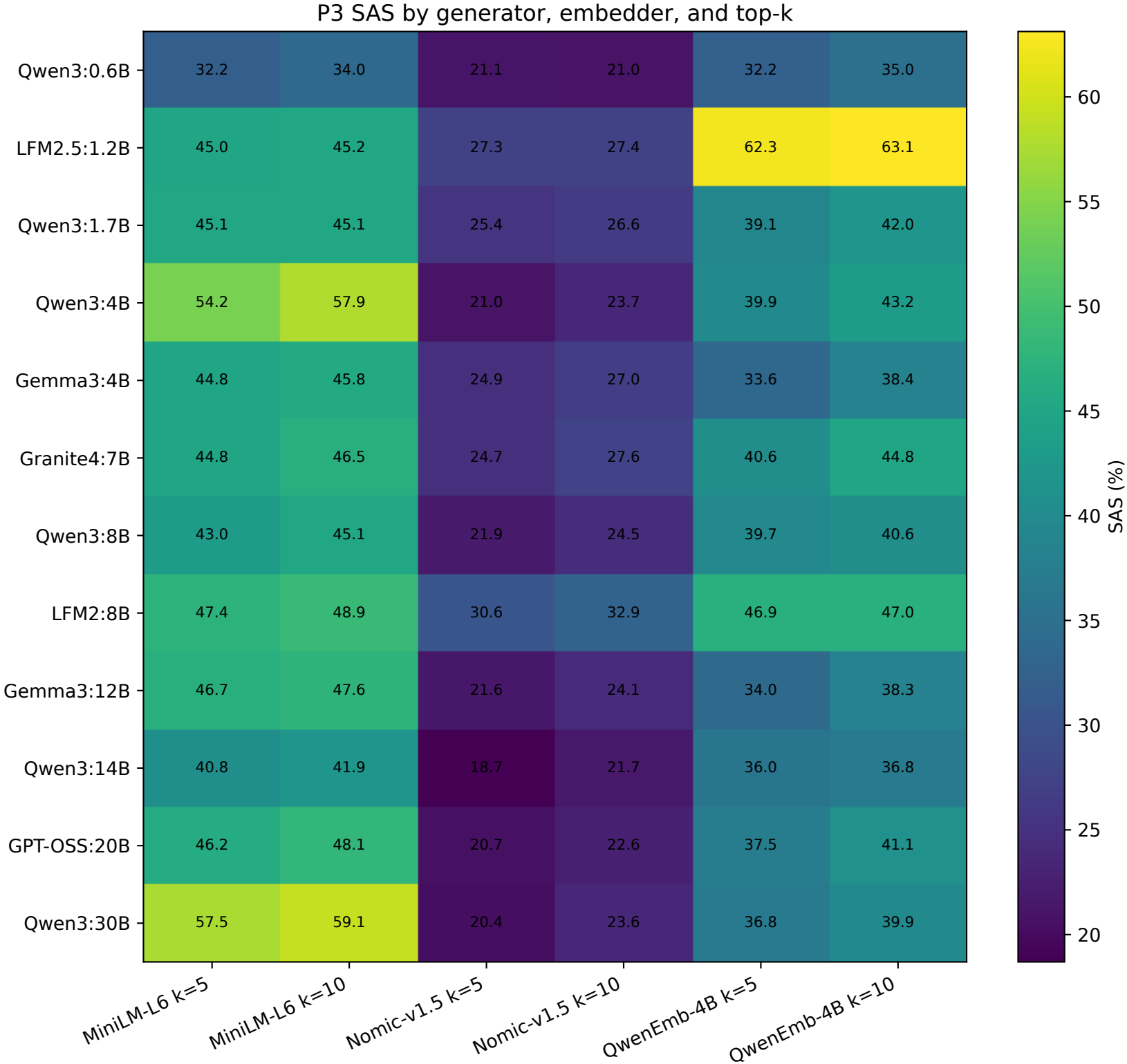

Runtime failure count by generator and pipeline
Failed trials (count)
0
5
10
15
20
25
30
35
P1
P2
P3
17
30
1
14
8
35
4
9
15
31
16
35
2
18
7
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B


Quality-latency trade-off
Best configuration F1 (%)
Mean total time per question (s)
P1
P2
P3
Qwen3:30B
Qwen3:4B
Gemma3:12B
Gemma3:4B
GPT-OSS:20B
Qwen3:1.7B
Qwen3:8B
Qwen3:14B
Granite4:7B
Qwen3:0.6B
LFM2:8B
LFM2.5:1.2B
Qwen3:30B
Qwen3:4B
Gemma3:12B
Gemma3:4B
GPT-OSS:20B
Qwen3:1.7B
Qwen3:8B
Qwen3:14B
Qwen3:0.6B
Qwen3:4B
Qwen3:14B
LFM2.5:1.2B
Qwen3:8B
Qwen3:4B
GPT-OSS:20B
Gemma3:12B
Qwen3:30B
Gemma3:4B
Granite4:7B
LFM2:8B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:0.6B
60
50
40
30
20
10
5
10
15
20
25
30


Generator scaling: semantic answer similarity (SAS)
P1
Best P2
Best P3
SAS (%)
0
10
20
30
40
50
60
20.0
35.4
35.0
28.2
66.1
63.1
32.1
49.3
45.1
36.4
61.5
57.9
31.9
51.6
45.8
34.0
50.0
46.5
38.3
46.3
45.1
36.4
50.6
48.9
40.2
54.1
47.6
38.2
43.0
41.9
42.0
48.9
48.1
48.3
64.8
59.1
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B


Semantic vs lexical answer quality
P1
P2
P3
Mean SAS (%)
Mean F1 (%)
60
55
50
45
40
35
30
25
20
10
20
30
40
50
LFM2.5:1.2B
Qwen3:30B
Qwen3:4B
Qwen3:30B
Gemma3:12B
LFM2:8B
LFM2.5:1.2B
Gemma3:4B
Granite4:7B
Qwen3:1.7B
GPT-OSS:20B
LFM2:8B
GPT-OSS:20B
Qwen3:8B
Qwen3:4B
Gemma3:12B
Qwen3:30B
Qwen3:8B
Granite4:7B
Qwen3:14B
Qwen3:1.7B
LFM2:8B
Qwen3:4B
Qwen3:8B
GPT-OSS:20B
Gemma3:12B
Granite4:7B
Qwen3:14B
Qwen3:1.7B
Gemma3:4B
LFM2.5:1.2B
Qwen3:0.6B


P2 SAS by generator, embedder, and top-k
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B
MiniLM-L6 k=5
MiniLM-L6 k=10
Nomic-v1.5 k=5
Nomic-v1.5 k=10
QwenEmb-4B k=5
QwenEmb-4B k=10
31.7 30.7 23.0 21.6 35.4 33.4
62.3 60.8 49.2 47.2 63.4 66.1
48.1 48.2 32.7 34.6 48.2 49.3
58.7 60.3 38.2 42.2 60.7 61.5
51.2 49.5 32.4 33.4 51.6 50.6
46.7 47.2 34.5 36.6 48.5 50.0
43.7 46.3 30.6 31.7 44.4 45.8
48.9 50.6 38.1 40.7 47.1 48.7
51.5 50.6 34.6 35.8 52.5 54.1
41.7 42.4 28.5 30.5 41.5 43.0
48.9 48.0 33.5 35.5 47.3 48.5
61.1 64.8 41.0 43.7 61.5 64.7
SAS (%)
65
60
55
50
45
40
35
30
25


P3 SAS by generator, embedder, and top-k
Qwen3:0.6B
LFM2.5:1.2B
Qwen3:1.7B
Qwen3:4B
Gemma3:4B
Granite4:7B
Qwen3:8B
LFM2:8B
Gemma3:12B
Qwen3:14B
GPT-OSS:20B
Qwen3:30B
MiniLM-L6 k=5
MiniLM-L6 k=10
Nomic-v1.5 k=5
Nomic-v1.5 k=10
QwenEmb-4B k=5
QwenEmb-4B k=10
32.2 34.0 21.1 21.0 32.2 35.0
45.0 45.2 27.3 27.4 62.3 63.1
45.1 45.1 25.4 26.6 39.1 42.0
54.2 57.9 21.0 23.7 39.9 43.2
44.8 45.8 24.9 27.0 33.6 38.4
44.8 46.5 24.7 27.6 40.6 44.8
43.0 45.1 21.9 24.5 39.7 40.6
47.4 48.9 30.6 32.9 46.9 47.0
46.7 47.6 21.6 24.1 34.0 38.3
40.8 41.9 18.7 21.7 36.0 36.8
46.2 48.1 20.7 22.6 37.5 41.1
57.5 59.1 20.4 23.6 36.8 39.9
SAS (%)
60
55
50
45
40
35
30
25
20

Pipeline	Error stage	Count
P3	rewrite	82
P3	generate	58
P2	generate	51
P1	generate	51

Generator	P1 EM	P2 EM k = 5	P2 EM k = 10	P3 EM k = 5	P3 EM k = 10
Qwen3:0.6B	0.002	0.082	0.066	0.068	0.068
LFM2.5:1.2B	0.018	0.072	0.106	0.040	0.060
Qwen3:1.7B	0.002	0.154	0.166	0.120	0.150
Qwen3:4B	0.068	0.388	0.366	0.318	0.312
Gemma3:4B	0.130	0.244	0.210	0.212	0.186
Granite4:7B	0.008	0.088	0.070	0.070	0.064
Qwen3:8B	0.006	0.074	0.074	0.078	0.074
LFM2:8B	0.006	0.016	0.014	0.020	0.016
Gemma3:12B	0.190	0.286	0.276	0.234	0.212
Qwen3:14B	0.014	0.054	0.048	0.056	0.042
GPT-OSS:20B	0.088	0.220	0.206	0.188	0.186
Qwen3:30B	0.180	0.448	0.460	0.418	0.430

Generator	P1 F1	P2 F1 k = 5	P2 F1 k = 10	P3 F1 k = 5	P3 F1 k = 10
Qwen3:0.6B	0.075	0.221	0.198	0.198	0.206
LFM2.5:1.2B	0.115	0.130	0.171	0.095	0.112
Qwen3:1.7B	0.083	0.340	0.350	0.304	0.296
Qwen3:4B	0.186	0.510	0.480	0.433	0.432
Gemma3:4B	0.262	0.429	0.402	0.369	0.360
Granite4:7B	0.147	0.255	0.240	0.233	0.212
Qwen3:8B	0.127	0.279	0.281	0.267	0.273
LFM2:8B	0.124	0.208	0.200	0.203	0.194
Gemma3:12B	0.340	0.464	0.466	0.415	0.416
Qwen3:14B	0.133	0.255	0.259	0.240	0.240
GPT-OSS:20B	0.224	0.396	0.386	0.361	0.364
Qwen3:30B	0.339	0.564	0.584	0.524	0.534

Generator	P2 gEM-all k = 5	P2 gEM-all k = 10	P3 gEM-all k = 5	P3 gEM-all k = 10
Qwen3:0.6B	0.042	0.050	0.040	0.044
LFM2.5:1.2B	0.038	0.068	0.024	0.038
Qwen3:1.7B	0.088	0.106	0.070	0.086
Qwen3:4B	0.226	0.256	0.160	0.192
Gemma3:4B	0.138	0.142	0.124	0.116
Granite4:7B	0.044	0.034	0.036	0.030
Qwen3:8B	0.040	0.038	0.032	0.042
LFM2:8B	0.006	0.006	0.004	0.008
Gemma3:12B	0.176	0.190	0.132	0.124
Qwen3:14B	0.026	0.024	0.026	0.024
GPT-OSS:20B	0.112	0.132	0.092	0.096
Qwen3:30B	0.278	0.322	0.226	0.262

Generator	P2 gEM-hit k = 5	P2 gEM-hit k = 10	P3 gEM-hit k = 5	P3 gEM-hit k = 10
Qwen3:0.6B	0.082	0.066	0.068	0.068
LFM2.5:1.2B	0.072	0.106	0.040	0.060
Qwen3:1.7B	0.154	0.166	0.120	0.150
Qwen3:4B	0.388	0.366	0.318	0.312
Gemma3:4B	0.244	0.210	0.212	0.186
Granite4:7B	0.088	0.070	0.070	0.064
Qwen3:8B	0.074	0.074	0.078	0.074
LFM2:8B	0.016	0.014	0.020	0.016
Gemma3:12B	0.286	0.276	0.234	0.212
Qwen3:14B	0.054	0.048	0.056	0.042
GPT-OSS:20B	0.220	0.206	0.188	0.186
Qwen3:30B	0.448	0.460	0.418	0.430

Generator	P2 gF1-all k = 5	P2 gF1-all k = 10	P3 gF1-all k = 5	P3 gF1-all k = 10
Qwen3:0.6B	0.133	0.144	0.113	0.140
LFM2.5:1.2B	0.068	0.108	0.058	0.068
Qwen3:1.7B	0.197	0.236	0.164	0.171
Qwen3:4B	0.299	0.330	0.217	0.250
Gemma3:4B	0.250	0.279	0.207	0.216
Granite4:7B	0.142	0.148	0.122	0.125
Qwen3:8B	0.166	0.185	0.143	0.171
LFM2:8B	0.121	0.131	0.107	0.121
Gemma3:12B	0.277	0.323	0.219	0.239
Qwen3:14B	0.150	0.173	0.142	0.157
GPT-OSS:20B	0.213	0.260	0.185	0.206
Qwen3:30B	0.346	0.408	0.276	0.317

Generator	P2 gF1-hit k = 5	P2 gF1-hit k = 10	P3 gF1-hit k = 5	P3 gF1-hit k = 10
Qwen3:0.6B	0.221	0.198	0.198	0.205
LFM2.5:1.2B	0.129	0.171	0.095	0.108
Qwen3:1.7B	0.339	0.350	0.303	0.295
Qwen3:4B	0.509	0.480	0.433	0.432
Gemma3:4B	0.429	0.402	0.367	0.360
Granite4:7B	0.253	0.239	0.231	0.211
Qwen3:8B	0.278	0.281	0.266	0.273
LFM2:8B	0.208	0.200	0.201	0.192
Gemma3:12B	0.464	0.466	0.413	0.415
Qwen3:14B	0.255	0.259	0.240	0.240
GPT-OSS:20B	0.395	0.386	0.359	0.364
Qwen3:30B	0.563	0.584	0.522	0.533

Generator	P2 gSAS-all k = 5	P2 gSAS-all k = 10	P3 gSAS-all k = 5	P3 gSAS-all k = 10
Qwen3:0.6B	0.21300571696843976	0.24180002441266324	0.17756454892586307	0.23393945694950066
LFM2.5:1.2B	0.36505030311457815	0.4350511317104101	0.31150702845677736	0.37181401398591696
Qwen3:1.7B	0.2793380517389132	0.3354075131844355	0.23796867747162043	0.26839736050928037
Qwen3:4B	0.3656060007698834	0.4323735716771409	0.274026312212643	0.338191648538342
Gemma3:4B	0.30576021609269083	0.35546443694643676	0.24945546635054053	0.27106714716367425
Granite4:7B	0.2718932058420032	0.32296049230359497	0.2302200211081654	0.26705700415186584
Qwen3:8B	0.2646281277194949	0.31434645258585914	0.23258156580944556	0.2866896466157469
LFM2:8B	0.2735592011604458	0.32619255493022503	0.25408020434714856	0.3052541932482272
Gemma3:12B	0.31812783309072257	0.3818552829027176	0.25021394795551893	0.28145031003840265
Qwen3:14B	0.24831285199686376	0.2968146907173112	0.23456195597772045	0.27343561086368934
GPT-OSS:20B	0.27775701299496	0.3359731676532949	0.24645216182777377	0.28577429214169986
Qwen3:30B	0.3792115779889067	0.46069152854255824	0.30470535537612464	0.3564191500972994

Generator	P2 gSAS-hit k = 5	P2 gSAS-hit k = 10	P3 gSAS-hit k = 5	P3 gSAS-hit k = 10
Qwen3:0.6B	0.3532755567294682	0.33393013238626706	0.3212316811703084	0.3489148845153329
LFM2.5:1.2B	0.6270574508849531	0.649789640057832	0.6053791339322925	0.6192393983528018
Qwen3:1.7B	0.48010200288438476	0.49180239130813913	0.4460384989361341	0.4483474143450177
Qwen3:4B	0.6069309284351766	0.6150424399648804	0.5412958446164507	0.5770260513030395
Gemma3:4B	0.5154050697796047	0.505128559326753	0.44519720853306355	0.4568360708802938
Granite4:7B	0.479599505931139	0.49139541257172825	0.4348657020032406	0.4512351144682616
Qwen3:8B	0.443806250313563	0.4620355751477182	0.4281734823108171	0.4486851137052111
LFM2:8B	0.4780267033409327	0.495400495423004	0.4657656183540821	0.48259992308728394
Gemma3:12B	0.5241211999151856	0.5402747948393226	0.4637745559066534	0.47461655619740484
Qwen3:14B	0.4149540318960029	0.4301197030665521	0.4058524567125873	0.41818483111446714
GPT-OSS:20B	0.4876350523458876	0.4851889904772102	0.46050113768062806	0.48069269318027374
Qwen3:30B	0.6142004721135979	0.6471957436804059	0.5723031841364049	0.5896428304216571

Generator	P1 SAS	P2 SAS k = 5	P2 SAS k = 10	P3 SAS k = 5	P3 SAS k = 10
Qwen3:0.6B	0.19987362002333006	0.3536909353921194	0.334159328975162	0.32247506111638896	0.35015210651637996
LFM2.5:1.2B	0.282089251358062	0.6344975545406342	0.6606531109400093	0.6229335919804871	0.6312411507237703
Qwen3:1.7B	0.3212611830295868	0.48158866087214114	0.4925627332689413	0.45138703767232913	0.45092521013378434
Qwen3:4B	0.36426577166654167	0.6073402654640376	0.6152688750368823	0.5424132382564516	0.5788725131323826
Gemma3:4B	0.3188156384937465	0.5158118809312582	0.5058122207559645	0.4477947376891971	0.4583352766595781
Granite4:7B	0.3404812084082514	0.48496792577393355	0.49987210500612855	0.4481396491285414	0.46503445334360005
Qwen3:8B	0.3834091450677947	0.4444271312067117	0.46276020939461887	0.42993684348584427	0.4512775147533985
LFM2:8B	0.3637901364825666	0.48938271427340807	0.5059916322436183	0.47373478681966663	0.48867507273703814
Gemma3:12B	0.4024766840673983	0.5245251580178738	0.5405003242902457	0.4672707122657448	0.47558751633018254
Qwen3:14B	0.3817162018629812	0.41665879160516667	0.43034568451016486	0.40762402729667496	0.4186348318247878
GPT-OSS:20B	0.42035961446346215	0.48915394046811395	0.48541588125979035	0.46217374835125724	0.4812361885435014
Qwen3:30B	0.48273392463379716	0.6146089171740156	0.6476215146098947	0.5746151961102515	0.5914093444716013